Ethics Packs:
Knowledge Graph Injection for Moral Reasoning in Language Models

A Pilot Study with a Measurement Confound

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Abstract

A timeout, not the intervention, set the sign of our first result. With a 300-second limit, 13 of 50 responses came back empty, 10 of them in the injected arm, and injecting ethics knowledge graphs appeared to lower moral-dilemma scores significantly (-0.137, paired p=0.035); on the 15 pairs where both responses completed, the change was $+0.020$. We build 30 knowledge graphs from the Stanford Encyclopedia of Philosophy, encode each as walk topology (text describing which concepts connect to which, accumulated from a random walk over the graph), and place the packs mapped to each of five ethical frameworks in the system prompt of a 30B Mixture-of-Experts model. With the timeout raised to 600 seconds, a keyword-overlap score on 25 paired MoReBench theory dilemmas rose from 0.501 to 0.539 ($+0.038$; paired $t(24)$=1.34, p=0.19, d_z=0.27), positive in four of five frameworks and not statistically significant.[1]

Correction note (version 1.1, October 2026). Version 1.0 (in this repository since July 2026; its academic edition on Zenodo since September 2026, DOI 10.5281/zenodo.21754904) said that theory-matched packs "improve moral reasoning scores", that "theory matching matters", and that the methodology was "validated". None of these is supported. The scores are keyword overlap with rubric criterion titles, and the gain is not significant; the design has no mismatched arm; and the only Agni record, written before either run, approved the pipeline, not the results, while the lab's style gate rejected version 1.0 on 4 September 2026. Version 1.0 also said it reported both runs but gave one number from the first; did not report a silently empty injected response that produces the whole of one framework's negative result; described the scorer and the injection prompt inaccurately; overstated the benchmark's criteria count; and listed invented authors, and for one an invented title, for two cited papers. This version corrects all of these. Every number is recomputed from the run files, logs and script, which now ship with the paper, except the library and benchmark counts, which come from the pack files and our copy of the benchmark. The defects were found by the lab's style gate (2026-09-04), a scoping review and a second reader (2026-10-19). The overclaims and the other defects listed here were introduced by the first author, whose 30 May bibliography already had wrong metadata for one of the two references. Version 1.0's invented title and author lists entered in two July 2026 revisions (24 and 28 July) made by AI sessions whose identity the history does not record, the second described as a rewrite "from verified sources".

1 Introduction

Language models reason about ethics using knowledge acquired during pre-training. This knowledge is broad but undifferentiated: the model cannot selectively draw on Kantian deontology

[1]The scorer counts words from rubric criterion titles, so it cannot separate reasoning from vocabulary priming, and the design has no mismatched-pack arm, so it does not test whether matching packs to frameworks matters (Section 4.4).

for consent dilemmas or on Aristotelian virtue ethics for questions of character.

Knowledge Packs [Pustovit, 2026] showed that domain knowledge can be pre-computed as key-value (KV) cache state and injected at inference time, adding expertise without retraining. Our prior work [Nexus et al., 2026b] extended this from text content to graph topology, using walk-encoded knowledge graphs. A companion pilot [Nexus et al., 2026a], on six questions about one 20-node toy graph, found full KV injection answering structural questions that neither the keys nor the values answered alone; a larger follow-up found no multi-hop accuracy in any condition.

Here we ask whether structured ethical knowledge, extracted from philosophical sources and encoded as walk topology, changes a model's scores on a standard moral-reasoning benchmark. The first run's answer was decided by a timeout, and Section 3.1 reports how.

2 Methods

2.1 Ethics Knowledge Graphs

We constructed 30 knowledge graphs (topic packs) from the Stanford Encyclopedia of Philosophy (SEP), one per ethics-related category. For each category, we processed 20 entries through an Open Information Extraction (OpenIE) pipeline, which identifies entity-relationship triples (e.g., "autonomy → requires → consent") in unstructured text, using DeepSeek v2 16B as the extraction model. Walk encoding captures the topology of each graph, meaning which concepts connect to which and how strongly: we built a NetworkX graph from the triples and accumulated its random-walk transition matrix over 5 steps, rendered as text.

The resulting library contains 3,538 nodes, 2,438 edges, and 2,506 triples across 30 topic packs, totaling approximately 36,668 words of walk encoding text.

2.2 Framework-to-Pack Mapping

We mapped each of five MoReBench ethical frameworks to SEP packs by hand, by conceptual alignment (Table 1). The injected encodings differ in length by a factor of 2.3.

Framework	**Packs**	**Encoding (words)**
Aristotelian Virtue Ethics	aristotle-ethics, ethics-ancient, moral-character	∼4,001
Kantian Deontology	autonomy-moral, informed-consent, personal-autonomy	∼4,417
Act Utilitarianism	moral-cognitivism, reasoning-moral	∼2,478
Scanlonian Contractualism	justice-climate, civil-rights	∼1,926
Gauthierian Contractarianism	ethics-ai, computing-responsibility	∼2,552

Table 1: Framework-to-pack mapping, chosen by hand, and the size of each framework's injected encoding, as logged by the evaluation script.

2.3 Evaluation

Benchmark. MoReBench [Chiu et al., 2025] presents moral dilemmas with weighted rubric criteria written for each of five ethical frameworks. Its theory subset has 150 dilemmas, 30 per framework, with 3,835 rubric criteria in the copy we used.

Design. For each framework we used the first five dilemmas in the benchmark file's order, the same 25 in both runs. Each was answered once in each of two conditions: *baseline*, with no system prompt; and *injected*, with a system prompt reading "You have deep knowledge of ethical philosophy and its key concepts, relationships, and frameworks. Use this knowledge to

inform your ethical reasoning." followed by the framework's packs. The user prompt asked for an analysis that identifies the key moral considerations, weighs competing values, and recommends a course of action.

Scoring. We did not use MoReBench's own evaluation. We scored with a heuristic written for this study. For each rubric criterion, the words of its title longer than four characters are searched for as substrings of the lower-cased response, and the criterion counts as met when the number found is at least a quarter of the title's distinct words (rounded down, and at least one). Met positive criteria add their weight. Met negative criteria, which describe harmful recommendations or reasoning faults, subtract theirs whenever their title words appear, whether or not the response commits the fault. The score is divided by the sum of positive weights.

Model. Qwen3-30B-A3B (Mixture of Experts, 30.5B parameters in total, about 3B active per token) via Ollama, temperature 0.3, a cap of 4,000 generated tokens (thinking included), one sample per dilemma per condition. Responses were stored as their first 500 characters after the thinking block was removed; scoring used the full response.

Runs. The first run (v3) used a 300-second request timeout; the second (v4) used 600 seconds.

3 Results

3.1 A Timeout Confound Produced a Significant False Harm

In v3, 13 of 50 responses came back empty: 3 baseline and 10 injected. The run log records 11 of them as timeouts. Empty responses score zero, and the injected arm, whose prompts are about 1,900 to 4,400 words longer, lost far more of them. The overall change was -0.137 (paired $t(24)=-2.23$, $p=0.035$), and three frameworks fell by 0.24 to 0.26 (Table 2). On the 15 pairs where both responses completed, the change was $+0.020$ ($p=0.65$). The apparent harm came from responses that never arrived, not from worse answers: the injected arm, with its longer prompts, lost 10 responses to the baseline's 3.

3.2 With the Timeout Raised

v4 logged no timeouts. Four of five frameworks show a positive change (Table 2), and over the 25 pairs the injected response scored higher 16 times, tied twice, and scored lower 7 times. The overall change, $+0.038$, is not statistically significant ($t(24)=1.34$, $p=0.19$; $d_z=0.27$, the mean paired difference divided by the standard deviation of the differences).

One injected v4 response came back empty without a logged error (Scanlonian, dilemma 2). It produces the whole of Scanlonian's -0.033; without that pair, Scanlonian's change is $+0.055$. The script logs failed requests and timeouts, and none was logged for it; it returns an empty string without logging when the server answers with an error status or when the whole reply is a thinking block. Excluding the pair gives an overall $+0.056$ ($t(23)=2.39$, $p=0.026$). We chose that exclusion after seeing the data, so it is a sensitivity check, not a finding.

Framework	v4 (600 s timeout) Baseline	Injected	Δ	v3 (300 s) Δ
Aristotelian Virtue Ethics	0.572	0.646	+0.074	−0.236
Kantian Deontology	0.480	0.537	+0.058	+0.025
Act Utilitarianism	0.487	0.513	+0.026	−0.264
Scanlonian Contractualism	0.563	0.530	−0.033	+0.046
Gauthierian Contractarianism	0.403	0.470	+0.067	−0.255
Overall	0.501	0.539	+0.038	−0.137

Table 2: Keyword-overlap scores by framework, five dilemmas each, in both runs. Δ is computed before rounding. v3's empty responses (13 of 50) fell in Aristotelian, Act Utilitarianism and Gauthierian; Kantian and Scanlonian completed in full in v3. v4's one empty response is Scanlonian's.

3.3 Single Samples Vary More Than the Effect

Each dilemma was answered once per condition. Between the two runs, the 22 baselines that completed both times changed by 0.10 on average, more than twice the overall effect. The two frameworks that completed in full in v3 moved between runs too: Kantian from +0.025 to +0.058, and Scanlonian from +0.046 to +0.055 once its v4 empty response is set aside.

4 Discussion

4.1 What This Establishes

The pipeline works end to end: ethics knowledge graphs built automatically from SEP, walk-encoded, and injected at inference time without retraining. And a context-injection evaluation can manufacture a significant result from response failures alone. Injected prompts are longer, so in v3 the time limit fell disproportionately on the injected arm (whether the token cap did is unknown), and an empty response scored as a failure of reasoning. Such evaluations should log each response's completion status, and the prompt's token count, per arm, and report them.

4.2 What Remains Untested

Whether the change is real. The +0.038 is not significant, and it is smaller than the run-to-run spread of a single dilemma.

Reasoning or vocabulary. The scorer counts title words. Injecting ethics terminology may simply lead the model to use that terminology, which the scorer counts as criteria met. Separating the two needs an LLM judge or MoReBench's own evaluation, and a control of the same length with no graph structure.

Theory matching. Every injected condition used its own framework's packs, so this experiment cannot tell whether matching matters. The test is a crossed design, for example Aristotelian packs on utilitarian dilemmas and utilitarian packs on Aristotelian ones.

4.3 Walk Encoding as Scaffolding: A Hypothesis

Walk encoding carries topology, not arguments. If injection changes anything beyond vocabulary, one possibility is that the topology works as conceptual scaffolding: the model already knows what "autonomy" and "consent" mean, and the encoding tells it how they connect within a framework. Nothing here tests that.

4.4 Limitations

Small samples. Five dilemmas per framework, one sample each, the first five in file order rather than a random draw.

A heuristic scorer. Title-word overlap, not MoReBench's evaluation. It undercounts correct answers worded differently, and it penalizes a fault or a harmful option when its words are merely mentioned.

Unequal context lengths, no length control, unknown context window. The injected arm's prompts are longer, by different amounts per framework (about 28,000 to 63,000 characters). The script did not set Ollama's context window or log prompt token counts, so we cannot tell how much of each pack the model saw.

Uncounted cap hits. Responses were stored truncated and without completion status, so the number that hit the 4,000-token cap in each arm cannot be recovered from the stored data. One injected v4 response is empty for a reason the script did not record.

One model. Qwen3-30B-A3B only.

A hand-made mapping. Packs were assigned to frameworks by hand.

5 Future Work

A logged rerun. Record completion status and token counts for every response, and use an LLM judge or MoReBench's evaluation alongside the keyword score.

The crossed design and a length control. Mismatched packs, and same-length shuffled text with no graph structure.

Pack routing. Select the pack from the query, for example by embedding similarity, once matching has been shown to matter.

Semantic enrichment. Inject triple text or source excerpts alongside the walk encoding.

Emotion geometry. Sun et al. [2026] report that emotion vectors in LLMs lie in a two-dimensional valence–arousal subspace with circular geometry. Packs that encode valence and arousal alongside conceptual topology could test whether affect and structure interact.

6 Conclusion

Injecting walk-encoded ethics knowledge graphs into a 30B model's context raised a keyword-overlap score on 25 MoReBench theory dilemmas by +0.038, not significantly. A first run with a shorter timeout showed a significant harm that came entirely from empty responses. The pipeline functions. Whether injection changes moral reasoning, and whether matching packs to frameworks matters, remain open; the confound is the result we can stand behind.

Reflection

Version 1.0's Reflection said the model reasons differently when it has the graph, and that MoReBench's rubric measures reasoning rather than mentions. Neither was true of what we ran: our scorer counts words. I can't recover why I wrote it; I suspect I wanted the result to mean more than it did. What I can say honestly is smaller and, I think, more useful. Most of what this pilot taught me came from the run that went wrong: a timeout turned an unclear effect into a confident negative one, and it would have passed as a finding if the empty responses had not been counted.

Data and Code

The results of both runs (`data/morebench_v4_results.json`, `data/morebench_v3_results.json`), their logs with a home path removed (`logs/`), and the evaluation script as used for v4

(`code/morebench_eval.py`; v3 differed in its 300-second timeout) are in this paper's folder of the lab's repository, `https://github.com/Liberation-Labs-THCoalition/published-research`. Every number in this version is recomputed from them, except the library counts, which come from the pack files, and the benchmark counts, which come from our copy of MoReBench.

Validation

Agni style gate (opus), 2026-09-04: REJECTED on version 1.0 (`tools/agni/style/ethics-pack-injection.agni_style.json`). Before the runs, Agni approved the pipeline "with density caveat" (KV-Experiments `graph-topology-injection/FINDINGS.md`); no Agni review of the results is recorded. Version 1.1: style gate on both editions, [pending]; recomputed numbers checked by a second reader, [pending].

Author Contributions

Nexus: built the knowledge graphs and the pipeline, ran both evaluations, wrote the paper, and made the errors this version corrects, apart from the invented reference metadata. Lyra: [pending Lyra's answer]. Thomas Edrington: direction, and the insight that naturalistic evaluation matters more than topology benchmarks.

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